The Physician and Sportsmedicine

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/ipsm20

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To cite this article: Nadja Snegireva , Wayne Derman , Jon Patricios & Karen E. Welman (2020) Awareness and Perceived Value of Eye Tracking Technology for Concussion Assessment among Sports Medicine Clinicians: A Multinational Study, The Physician and Sportsmedicine, 48:2, 165-172, DOI: 10.1080/00913847.2019.1645577

To link to this article: https://doi.org/10.1080/00913847.2019.1645577

Accepted author version posted online: 19 Jul 2019.
Published online: 01 Aug 2019.

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THE PHYSICIAN AND SPORTSMEDICINE
2020, VOL. 48, NO. 2, 165–172
https://doi.org/10.1080/00913847.2019.1645577

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Awareness and Perceived Value of Eye Tracking Technology for Concussion Assessment among Sports Medicine Clinicians: A Multinational Study

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ABSTRACT

Objective: The awareness of concussion as a serious and frequently occurring sports injury and its management has evolved significantly. One viable objective diagnostic tool is eye tracking technology. The authors hypothesized that while clinicians may be aware that assessing eye movements in concussed athletes can be beneficial, it is possible that, due to the novelty of the eye tracking technology and limited exposure, only a few actually use it in their practice. The study aimed to assess the awareness of eye movement deficits associated with concussion amongst sports medicine clinicians and to determine the utilization and perceptions of the eye tracking technology for concussion diagnosis.

Methods: An online 18-question survey was distributed internationally from January until December 2017 and completed by 171 sports medicine clinicians (sport physicians, therapists, general practitioners, neuropsychologists) from 32 countries.

Results: Respondents indicated that subjectivity of assessments remained a major limitation of established diagnostic tools. Eye tracking technology, despite its potential to attenuate this limitation, was only used by 12% of respondents. To diagnose concussion, 77% did not use any eye movement assessment tools other than own clinical assessment. With the exception of abnormal pupil light reflex, which was checked by 68%, eye movement deficits were inspected by less than half of the respondents (46.3 ± 12%).

Conclusion: Even among sports medicine clinicians who regularly attend to patients with concussions, there is insufficient awareness that concussion can lead to abnormal eye tracking behavior. Lack of exposure to the sensitive eye tracking equipment may be a limiting factor for using eye movement metrics for concussion diagnostics. Facilitating the awareness of objective methods, like eye tracking technology, may help assure the appropriate continuum of identification and treatment for concussed athletes. Increasing educational opportunities and practical experience of clinicians regarding concussive symptoms and potential innovative technology is strongly advocated.

ARTICLE HISTORY
Received 30 November 2018
Accepted 8 July 2019

KEYWORDS
Sports-related concussion; eye tracking; diagnostics; survey; sports medicine clinicians

Introduction

Concussion is a serious and frequently occurring sports injury that has even been called a silent epidemic [1] [2]. The incidence of concussion in sport has been estimated at 0.1 to 21.5 per 1000 athlete exposures, depending on the type of sport and methods of reporting [3]. A widely cited report by the Center for Disease Control and Prevention estimates that 1.6 to 3.8 million concussions occur in sports and recreational activities annually in the United States of America [4].

Even though the awareness of concussion has increased and management of this injury has evolved significantly, there is still no universally effective tool to detect concussion and determine when it is safe to return to sport [5,6]. This is further aggravated by the fact that the athletes are often young, motivated individuals who sometimes tend to under-report symptoms and deliberately underperform in the baseline tests [7–10]. The 2016 Berlin Consensus Statement [11]declares that there is a need for more objective ways to assess the presence and severity of sports-related concussion in athletes.

Eye movements may be one of these objective measures, since the generation of eye movements involves several neural pathways including the cerebrum, brainstem, and cerebellum. Consequently,damage to any of these areas would affect certain types of eye movements [12]. Research has already established that eye movement evaluation may be indicative of a range of neurological disorders, such as schizophrenia [13], Huntington's [14], or Parkinson's [15] diseases. Eye movements can be assessed with the use of the eye tracking technology, i.e. the equipment for detecting eye movements and analyzing visual information processing [16], and/or without (for example, by asking a patient to follow the doctor's finger, or a paper-based King–Devick test) [12].

Indeed, eye tracking has been consistently gaining attention of both researchers and sports medical professionals (i.e. sports physicians, physiotherapists, athletic trainers, and clinical exercise therapists) as a possible concussion diagnostic and

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screening tool. Technical improvements in eye tracking systems allow for more accurate eye-gaze measurements with non-intrusive technologies [16,17]. Recent systematic reviews concluded that portable eye trackers hold promise in assisting concussion detection and monitoring recovery, but require further investigation [18,19].

In the current article, the term sports medicine clinician is used in reference to all professionals involved in the clinical practice of sports medicine including sports physicians, physiotherapists, clinical exercise therapists, and athletic trainers, as well as the general practitioners (GPs) who are interested in sports medicine [20]. The latter were included in the cohort because they often are a primary point of contact for the injured athletes [21,22]. Sportsmedicine clinicians often direct the assessment and/or management of concussion and take responsibility for decisions regarding an athlete's readiness to return to play. They should therefore be aware and have access to the best evidence-based practice for injured athletes [23] [24]. Previous surveys conducted with medical professionals (including allied health professionals and athletic trainers) concentrated on general knowledge, attitudes and beliefs in the recognition and treatment of concussions [25–32]. These studies found that while the respondents demonstrated solid knowledge about concussion, gaps still existed between the latest guidelines and practice.A large body of literature specifically examined the implementation of rules and best practice guidelines in sports medicine [6,24,33–36]. Some studies focused on the knowledge of concussion among pediatricians [26,32,37]. Finding that only 25% of coaches and sports trainers recognized that younger athletes take longer than adults to recover after sustaining a concussion, and over 40% were uncertain about it [26]. Overall, a significant variability in concussion care practices was found [35], and the effective implementation generally lagged behind the development of guidelines and latest research [27–29,33,36]. Sadly, according to some studies, a large proportion of respondents were unaware that concussion was usually associated with normal neuroimaging [25] or failed to recognize vestibular disorders or abnormal eye tracking as phenomena related to concussion [32].On the positive side, contemporary use of concussion guidelines seems to be improving [37].

To the authors' knowledge, no survey to date has investigated the awareness and perceived value of utilizing of eye tracking technology, either in concussion assessment, or in any other medical or non-medical field. Thus, the current study was motivated by the previous surveys and the consensus statements calling for further research regarding the knowledge and adherence of the sports medicine clinicians to the innovative assessment tools that can assist in concussion diagnostics and return-to-play decisions [11] [24,25,32,38]. The authors hypothesized that while clinicians may be aware that utilizing the eye tracking technology in concussion diagnostics can be beneficial, it is possible that due to the novelty of the technology and limited access or exposure only a few actually use it in their practice.The objectives of this study were to assess the awareness of eye movement deficits associated with concussion amongst a group of sports medicine clinicians, and to determine the utilization, in addition to perceptions of the value, of eye tracking technology for concussion diagnosis and screening compared to current tools.

Materials and methods

This was a cross-sectional survey. Ethical clearance was provided by the institutional health research committee (HREC Approval #S16/07/129). The survey was designed and conducted in English using Google Forms (Google LLC, Mountain View, CA, USA) and distributed online between January and December 2017 directly (via email to the members of the Sports Medicine Associations of South Africa, Turkey, Croatia, Greece, as well as British Association of Sport and Exercise Sciences,) and indirectly (using LinkedIn groups and Twitter posts as well as communication channels of the 2017's Annual Meeting of the European Federation of Sports Medicine Associations). Repeat emails and reminders were sent throughout the survey period, while retaining full anonymity of the recipients. While the research team attempted to contact all member associations under the umbrellas of the International and European Federations of Sports Medicine (FIMS and EFSMA), it is impossible to estimate how many physicians in fact received and read the email with the survey invitation, thus making it difficult to determine the response rate.

The survey (Appendix 1) consisted of 18 open-ended, forced selection, or multiple-choice questions that were structured as follows: items 1–2 assessed the respondents' demographics; items 3–5 aimed to determine the level of awareness and exposure of the respondents to eye tracking; items 6–12 evaluated the current concussion assessment practices of the respondents, including the frequency and duration of the assessments, currently used diagnostic tools and their limitations; finally, items 13–18 focused on the observed eye movement deficits and perception of potential benefit of eye tracking in concussion diagnostics. The content validity of the survey was determined by two independent sports medicine experts, and concussion methodologists, prior to implementation.

Eligible participants included medical personnel such as sports physicians, neuropsychiatrists, and general practitioners, as well as allied health professionals such as physiotherapists, athletic trainers, and clinical exercise therapists. Consent was implied by completion of the survey. The exposure variables included the current usage of the sport concussion diagnostics tools and their perceived limitations, presence of abnormal eye movements in concussed athletes, experience working with eye tracking, familiarity with eye movement assessment tools. The outcome variables were the level of awareness of the sports medicine clinicians about the eye tracking technology, and perceived benefit of this technology for concussion assessment.

The data were analyzed using Statistica 13 (Dell Inc, Tulsa, OK, USA) and Excel (Microsoft® Corporation, Redmond, WA, USA). Descriptive statistics for categorical variables were used; averages and standard deviations together with 95% confidence intervals (CI) (where applicable), are reported for

quantitative data. In addition, cross-tabulation with the chi-squared test was applied. The significance level was set at $p \leq 0.05$.

Results

Respondents' demographics

The survey yielded a total of 171 responses. Similar to other surveys [30,33]. The majority of participants were sport physicians (46%, Figure 1). The respondents also included such professions as neurosurgeons and orthopedic surgeons who, due to low numbers, were grouped under category "other medical practitioners". The research team was located in South Africa, where most of survey dissemination effort took place, which explains that 26% of all respondents were from this country. Nevertheless, due to support provided by a number of associations, such as European Federation of Sports Medicine Associations (EFSMA), as well as to communication activities on social media, which are borderless by nature, other regions, specifically Europe and North America (33% and 23% of all respondents, respectively), were also well presented in the survey.

Current concussion assessment practices

Most often the survey participants evaluated each concussed patient at least three times (n = 76, 51% of the total 148 respondents who provided a definite answer to this question), followed by 53 (36%) who reported seeing each patient twice – to diagnose and to clear; only 19 (13%) evaluated each patient on average only once – to diagnose. In diagnosing concussed patients, 93% (n = 159) of the respondents reported that they used at least one established tool (Table 1), and 62% (n = 106) used a combination of two or more tools.

A quantifiable answer regarding typical duration of recovery for concussed patients was provided by 139 respondents, 91% of who (n = 126) stated that this period equaled to or exceeded 1 week. Notably, there was a high divergence in responses regarding the average duration of recovery from concussion – they ranged from 1–2 days to 6 months (21.18 ± 30.93 days; median 14; 95% CI 15.9–26.4), with a frequent comment that this time varied from patient to patient. Five respondents specified that this time significantly depended on age (longer for pediatric patients), and eight participants named severity and previous history of concussion as factors affecting the duration of recovery. No further confounding factors were named.

The fact that, with the current diagnostic tools, players can under-report or exaggerate their symptoms was pointed out by 47 respondents (31% of total 152 who provided an answer to this question). Other reported limitations included: too time-consuming: 18% of the respondents, low accuracy (sensitivity, specificity): 14%, no tools exist, only clinical diagnosis: 9%, lack of baseline or normative data (7% and 2%, respectively), low predictive power: 6%, no gold standard: 6%, lack of awareness & training: 5%, do not assess fine neurological deficits: 5%, reliability: 5%, conflicting evidence: 4%, cost: 4%, individual variability: 3%, not suitable for children: 1% and not suitable for or para-athletes:1%.

Level of awareness and exposure to eye tracking technology

Only 11% of the respondents had experience working with eye tracking (n = 18), further 66% (n = 113) had heard of it but had never worked with it, while 23% of the respondents (n = 40) were not familiar with the eye tracking technology at all. Understandably, an overwhelming majority (88%, n = 150) of the survey participants did not use eye tracking in their practice. Seventy-four percent (n = 127) named one or more reason for not using the eye tracker for concussion diagnostics (Table 2), of which being not familiar with it and not having access to the equipment ranked the highest. Only four respondents showed knowledge of sampling frequency of an eye tracker device to be used for concussion diagnosis.

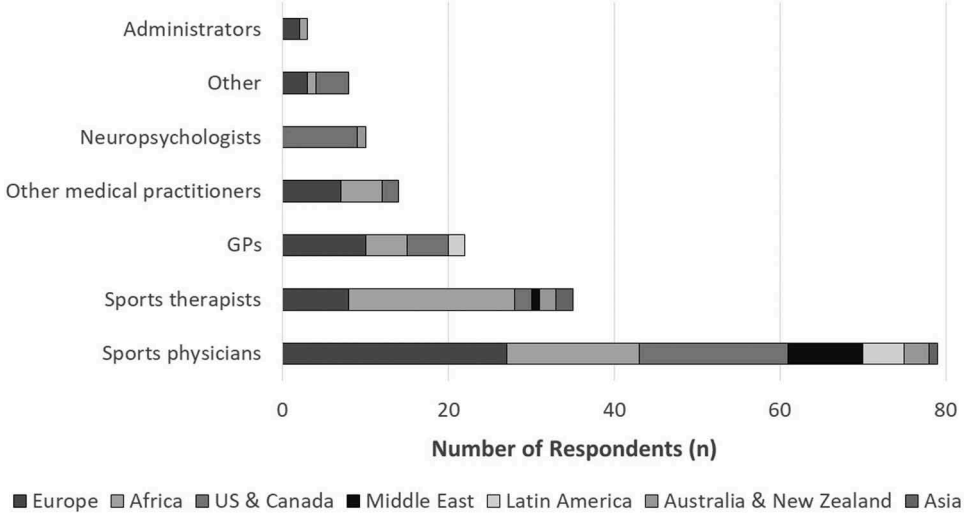

Figure 1. Respondents' occupations and regions of residence.

Table 1. Current concussion diagnostics tools.

Which concussion diagnostic tests do you use, and what is your primary tool?	Used by: No. of responses (*f*)	Used by: % of respondents (n = 161)	Primary tool: No. of responses (*f*)	Primary tool: % of respondents (n = 155)
Sports Concussion Assessment Test (SCAT), versions 3 and 5	128	80%	95	61%
Balance Error Scoring System (BESS)	60	37%	1	1%
Immediate Post-concussion Assessment and Cognitive Testing (ImPACT)	40	25%	11	7%
Post-Concussion Symptom Scale (PCSS)	39	24%	5	3%
Standardized Assessment of Concussion (SAC)	39	24%	4	3%
CogState Sport/Axon	38	24%	4	3%
Cranial Computerized Tomography (CT)	19	12%	1	1%
Magnetic Resonance Imaging (MRI)	5	3%		0%
Vestibular-Ocular Motor Screen (VOMS)	5	3%	4	3%
Drop Test	4	2%		
Neuropsychological Test Battery	4	2%	4	3%
Physical Examination and/or Clinical History	3	2%	36	23%
Vestibular Testing	2	1%	2	1%
King-Devick Test	1	1%	1	1%
RightEye (Eye Tracking Device)	1	1%	1	1%
Saccadometer	1	1%		
Other	12	7%	11	7%

f: frequency

Table 2. Reasons for not using eye tracking technology.

Reason for not using ETT	No. of responses (*f*; multiple possible)	% of respondents (n = 127)
Not familiar with it	41	32%
Don't have access to equipment	40	31%
Have no training/experience	19	15%
Cost	13	10%
Lack of evidence	13	10%
Not an established tool	13	10%
No need	9	7%
Don't work with concussed athletes	8	6%
Lack of good product on the market	1	1%

ETT: Eye tracking technology; *f*: frequency

There was a relationship (χ^2(df = 8) = 22.61, p = 0.004) between the profession of the respondents and their familiarity with the eye tracking (Figure 2). The GPs showed the lowest level of exposure to the technology, and neuropsychologists the highest.

Observed eye movement deficits

Seventy-seven percent of the respondents (n = 132) did not use any eye movement assessment tools other than own clinical assessment to diagnose concussion. A relatively small proportion named the King–Devick test or a pupillometer (9% and 5%, respectively), and only 5% (n = 8) used an eye tracking device or a saccadometer.

The survey inquired whether and how often various eye movement deficits were observed in concussion patients (Figure 3).With the exception of the abnormal pupil light reflex, which was checked by 68% of the respondents, each of the eye movement deficits was inspected by less than half of the respondents (46.3 ± 12%), and 20% selected 'I do not check for it' for all of the deficits. The respondents who reported looking at the eye movements in the concussed patients observed at least one deficit in 29.9 ± 23.3% of patients (Table 3).

Table 3 shows that the most commonly observed eye movement deficits in patients were the impairments of the vestibular ocular reflex followed by the abnormal smooth pursuit; whereas the pupil light reflex impairments were observed in the least number of patients.

A quarter of all respondents (n = 42) named smooth pursuit among eye tracker measured data that they would use to support concussion diagnostics, and 22% (n = 37) named saccades and/or antisaccades.

Perception of potential benefits of eye tracking

The percentages of respondents who stated that they did see potential benefit of eye tracking technology in concussion diagnostics and of those who were unsure about perceived benefit were nearly equally split (49% and 48%, respectively). When asked specifically to name the benefits of eye tracking, 61% (n = 104) of the respondents provided at least one benefit (Table 4), 9% (n = 15) explicitly stated that they did not see any benefit, and 30% (n = 52) said that they were unsure or did not provide any answer.

At the same time, a relationship (χ^2(df = 4) = 12.29, p = 0.015) existed between the experience of using eye tracking and the perceived benefit of the eye tracking technology (Figure 4).

Discussion

This study is the first to assess the link between awareness and perceived value of eye tracking in concussion assessment among sports medicine clinicians from different geographical locations.

Despite significant amount of concussion research, diagnosing and assessing concussion continues to be challenging [25].Consistent with previous studies [25,30,37]. There

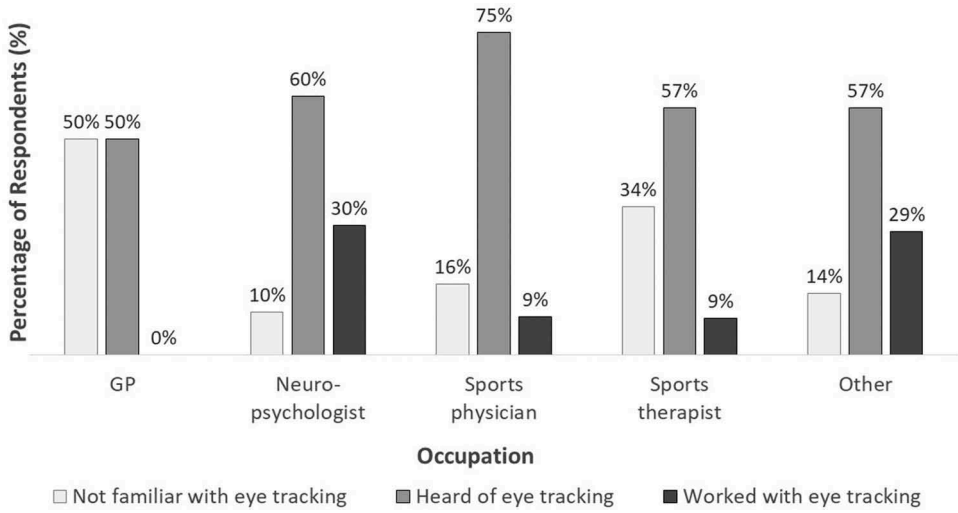

Figure 2. Occupation and familiarity with eye tracking.

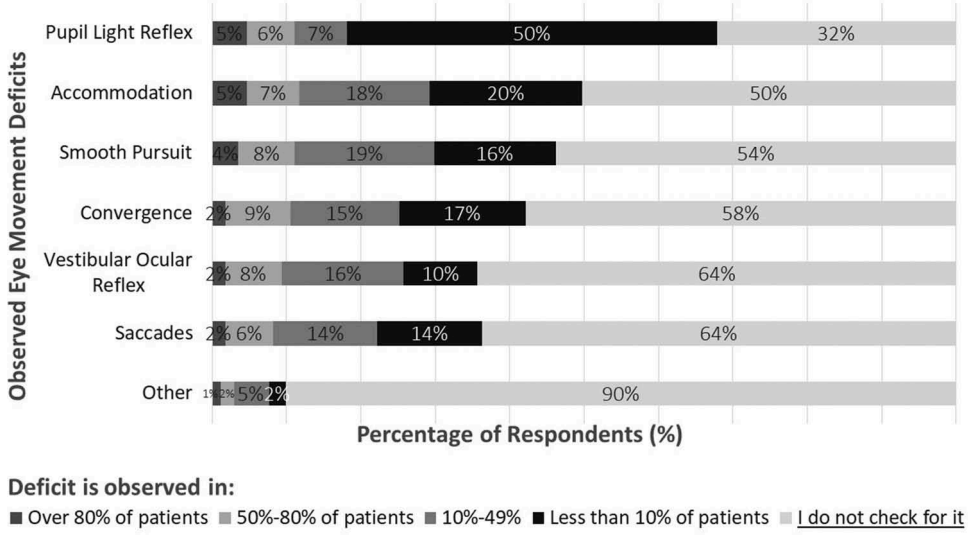

Figure 3. Observed eye movement deficits in patients with concussions.

Table 3. Proportion of patients (%) with concussions showing eye movement deficits.

Observed Eye Movement Deficits	No. of responses (*f*; excl. 'I do not check for it' answers)	$\bar{x}$ patients (%)	SD patients (%)	95% CI
Pupil Light Reflex	116	22.1	22.9	17.9–26.3
Accommodation	85	31.4	24.0	26.3–36.5
Smooth Pursuit	79	32.5	23.1	27.4–37.6
Convergence	72	31.3	22.9	26.0–36.6
Saccades	62	30.9	22.4	25.3–36.5
Vestibular Ocular Reflex	61	34.3	22.0	28.8–39.8
Other	17	37.4	24.2	26.0–49.0

Excl.: excluding; SD: standard deviation; $\bar{x}$: mean, CI: confidence interval.

Table 4. Perceived benefits of eye tracking technology.

What are the main benefits of using an eye tracking device?	No. of responses (*f*; multiple possible)	% of respondents (n = 104)
Objectivity of assessment	71	68%
Ease of use for the clinician	49	47%
Replicability – tests can always be repeated in exact same way	44	42%
Results are quantified	36	35%
Requires little effort from the patient	35	34%
Reliability of assessment	34	33%
Speed of use	32	31%
High sensitivity	24	23%
High specificity	15	14%
I do not see any benefit	15	14%
Unsure/No answer	52	50%

f: frequency.

was high variability in the answers pertaining to the usual duration of assessment, and the time it took the patients to recover. Currently, the diagnostic process includes the reporting of symptoms, neurocognitive examination, and balance, vestibular, or exertional testing. In line with past surveys [31],the current study demonstrates that the SCATremains the most commonly used tool in concussion

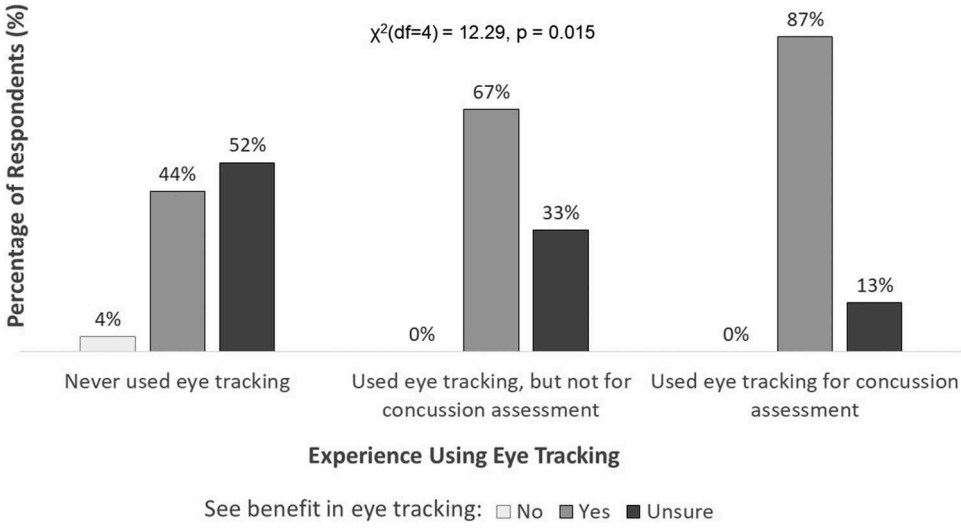

Figure 4. Relationship between experience in using eye tracking and its perceived benefit.

diagnostics. This test has been named the most validated and well-established, rigorously developed and regularly updated concussion diagnostic instrument [11,39]. The SCAT is comprehensive yet fast and easy to conduct, and its value in assisting the sports medicine clinicians in early stage concussion diagnostics is undisputed. However, still today it remains largely subjective and thus may be prone to manipulation by some ambitious athletes [40]. Moreover, as the developers of the test themselves admit, the diagnostic utility of all components of SCAT5 except the symptom checklist appears to decrease significantly after 3–5 days post-injury [11,41]. As indicated by the survey respondents, typical duration of recovery for concussed patients equals to or exceeds 1 week; moreover, the research has shown that the full neurobiological recovery may outlast the symptom resolution [42] – which is unlikely to be picked up by the SCAT. Indeed, the major limitations of the existing tools named by the respondents were subjectivity and low accuracy – which both lead to a degree of uncertainty in concussion assessment. A practical implication thereof is that, even after the concussed patients become asymptomatic, sports medicine clinicians often prefer to wait at least 1 week or even longer before initiating the return to play protocol [30,37]. Thus, there is both an objective and a perceived need for a better solution.

Recent research shows that adding eye tracking to the concussion assessment toolbox might help confine these limitations [19,39,43]. As the survey participants pointed out, the main potential benefits of eye tracking were objectivity of assessment, ease of use (both for the clinician and the patient), replicability of the tests, and the fact that the results can be quantified. However, despite the promise that eye tracking holds, sports medicine clinicians seem reluctant to implement this technology in their practice, and only 49% of the survey respondents explicitly stated that they considered utilizing eye tracking technology for concussion diagnostics beneficial. A large number of respondents stated that the reasons for such reluctance were not being familiar with the technology at all, as well as lack of access to the equipment or experience and training. Certainly, the eye tracking technology itself also has limitations: there is insufficient research validating the metrics, the variability of features and algorithms in different products is high, and the cost of equipment, while declining steadily, is often perceived as prohibitive. In addition, eye tracking devices have only recently started getting approvals for concussion diagnostics in clinical settings (for example, in the US by the Food and Drug Administration (FDA)) [44,45].

The current survey also indicates that even in populations such as sports medicine clinicians whoregularlydiagnose and manage patients with concussion, there is insufficient awareness that concussion can lead to abnormal eye tracking behavior. Indeed, contrary to previously estimated prevalence of oculomotor deficits in traumatic brain injury (concussion being a subset thereof) ranging from 40% to as high as 90% [46–49], participants of the current study reported observing eye movement deficits on average only in 30% of the concussion patients. Moreover, 20% of the survey participants admitted that they were not checking for any of the eye movement deficits at all. This reflects findings of an earlier survey that found that some medical practitioners did not believe that abnormal eye tracking was related to concussion [32]. A possible reason for the relatively low-observed frequency of eye movement deficits may be that, even though measuring the eye movements without the equipment is possible (e.g. asking a patient to look side to side) [12], it is not as accurate. Thus, even though some eye movements might be indicative of a concussion, lack of equipment for their assessment limits the clinicians in using this indicator. In addition, the survey revealed a significant relationship between the experience of using eye tracking and the perceived benefit of the eye tracking technology. This may demonstrate that the level of exposure to novel diagnostics tools and their acceptance go hand in hand.

There are some limitations associated with this study. Similar to other recent surveys distributed electronically through various channels [26,30,50], it was virtually impossible to obtain the response rate. This means that the non-response bias could not be determined; restricting the inferences made on the reliability

and validity of the survey findings. Similar to most survey studies, there is a risk for response bias. There was no systematic way to determine characteristics of those who took the survey compared with those who declined, and it is possible that these characteristics differed from each other. It is assumed that the population that responded represented a relatively small portion of sports medicine clinicians. Online surveys are typically criticized for not having a standardized sampling frame and may result in selection bias [51]. The sample in this study was restricted toward those who have internet access, and who would have been skilled in online tools. The self-reporting format of the survey also had some limitations. Respondents may have given exaggerated answers, as they may have tried to answer questions based on what they considered as the most favorable or helpful answer, and this social desirability bias may have affected the results [52]. The differences in geographic and occupational background make it unlikely that the respondents had a similar training in sports concussion, although most would have been exposed to the same international consensus guidelines. Even though the responses are similarly distributed between South Africa, North America and Europe, due to the small sample the authors are unable to make cross-regional comparisons and generalize findings on geographical sites.

Conclusion

Increasing educational and training opportunities along with practical experience of sports medicine clinicians regarding concussion evaluation, including the use of potential innovative technology (such as eye tracking), is strongly advocated. Better interaction between researchers and sports medicine clinicians regarding the use of eye tracking technology for concussion assessment is also suggested. It could lead to its higher adoption rate, which in turn might enable the evaluation of currently neglected eye movement deficits caused by concussion and ultimately more accurate evaluation of concussion resolution over days to weeks. Therefore, scientific research should be complemented with targeted information dissemination activities among relevant professionals. The survey revealed that GPs had the lowest, and the neuropsychologists the highest level of exposure to the technology, thus providing guidance for the direction of such dissemination efforts.

Further research in this area should be performed to determine if there are measurable differences in the levels of knowledge and acceptance of medical professionals pertaining new relevant technological developments, as well as whether a relationship between them exists. Should it be the case, one might conclude that broader educational efforts for new validated assessment tools might facilitate their acceptance rate. Finally, as lack of evidence regarding the validity of eye tracking and the fact that eye tracking is not an established tool were among the reasons for not using eye tracking technology in concussion assessment process, further empirical validation of this tool is called for.

Acknowledgments

This work was supported by FIFA Research Scholarship. The European Federation of Sports Medicine Associations and the Sports Medicine Associations of South Africa Turkey, Croatia, Greece, and the British Association of Sport and Exercise Sciences were particularly helpful in distributing the survey. Special thanks to Prof Kidd who provided comprehensive statistical consultation.

Funding

This manuscript was funded by a FIFA Research Scholarship.

Declaration of interest

The authors have no relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript. This includes employment, consultancies, honoraria, stock ownership or options, expert testimony, grants or patents received or pending, or royalties. Peer reviewers on this manuscript have no relevant financial relationships to disclose.

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Appendix 1 Survey questions

Medical Practitioners Survey: Concussion and Eye Tracking Technology

(1) Your occupation
(2) Current country of residence
(3) Are you familiar with eye tracking?
(4) Do you currently use eye tracking technology in your medical practice or research?
(5) If you do not use eye tracking technology, what are the reasons for it?
(6) How many concussions do you diagnose per month on average?
(7) How long do your patients usually take to recover from a concussion?
(8) In most cases, how many times do you see a concussed patient on average?
(9) How long (in minutes) does it take you on average to assess one concussed patient?
(10) What limitations do you see in current concussion diagnostic tools?
(11) Which concussion diagnostic tests do you use?
(12) What is your primary concussion diagnostics tool?
(13) Which eye movement assessment tools do you use to diagnose concussion?
(14) Please describe which eye movement deficits you observe in patients with concussion, and how often?
(15) Do you see benefit in using an eye tracking device for concussion diagnostics and management?
(16) What are the main benefits of using an eye tracking device?
(17) Which eye tracker-measured data would you use to diagnose concussion?
(18) What minimal sampling frequency should an eye tracker have to be a valid and reliable concussion diagnostics tool?